

1. Objectives:

1. To understand the concept of private access modifier, instance variables, scope, and this keyword in Java.
2. To learn how encapsulation works in java using OOP.

2. Introduction:

- **Encapsulation:** It is a process of capsulation and mapping the related variables and functions inside a class with proper security mechanisms for each variable or method. It also helps protect data by restricting direct access using access modifiers such as private. This improves data security and maintains proper control over class members.
- **Instance Variable:** It is a variable declared inside a class but outside methods or constructors. Each object of a class gets its own separate copy of instance variables, allowing different objects to store different values. Since private instance variables cannot be accessed directly from outside the class, Java uses **getter and setter** methods to provide controlled access to the data. A **getter method** is used to retrieve or display the value of a private variable, while a **setter method** is used to assign or modify the value of that variable. This process helps maintain encapsulation and data protection.
- **this keyword:** It is a special reference variable in Java that refers to the current object of a class. It is mainly used to distinguish instance variables from local variables when both have the same name. The **this** keyword can also be used to call constructors and methods of the current class.

3. Required Hardware and Software:

- **Hardware:** AMD RYZEN 7, RAM 8 GB
- **Software:** Visual Studio Code 1.109

4. Experiment Implementation:

Experiment 1: Use of Static keyword

```
package Static;
public class StaticMethod {
    int n1;
    int n2;
    static int add;
    StaticMethod(int a, int b) {
        n1 = a;
        n2 = b;
    }
    static int add(int a, int b) {
        return add = a + b;
    }
}

package Static;
public class Main {
    public static void main(String[] args) {

        int result = StaticMethod.add(10, 20);
        System.out.println("The result is: " + result);
    }
}
```

Experiment 1 output:

```
PS D:\BAUST CSE\Java Programing\7th_Lab> & 'C:\Program Files\Java\jdk-25.0.2\bin\
ezwan\AppData\Roaming\Code\User\workspaceStorage\4bed134accdc63fc9436ca55b0e3d438\
The result is: 30
PS D:\BAUST CSE\Java Programing\7th_Lab>
```

Experiment 2: Use of this keyword

```
package This;
public class Student {

    String name;
    int age;

    Student(String name, int age) {
        this.name = name;
        this.age = age;
    }

    public void show() {
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
    }
}

package This;
public class Main {

    public static void main(String[] args) {

        Student s = new Student("Paris", 20);
        s.show();
    }
}
```

Experiment 2 output:

- PS D:\BAUST CSE\Java Programing\7th_Lab> & 'C:\Program Files\Java\jdk-25.0.2\ezwan\AppData\Roaming\Code\User\workspaceStorage\4bed134accdc63fc9436ca55b0e3d
Name: Paris
Age: 20
- PS D:\BAUST CSE\Java Programing\7th_Lab>

Experiment 3: Use of getter and setter method

```
package SetGet;
public class Teacher {

    private String name;
    private String designation;

    public String getName() {
        return name;
    }

    public void setData(String name, String designation) {
        this.name = name;
        this.designation = designation;
    }

    public String getDesignation() {
        return designation;
    }
}

package SetGet;
public class Main {

    public static void main(String[] args) {

        Teacher t = new Teacher();
        t.setData("Zahim Sir", "Lecturer");

        System.out.println("Name: " + t.getName());
        System.out.println("Designation: " + t.getDesignation());
    }
}
```

Experiment 3 output:

```
● PS D:\BAUST CSE\Java Programing\7th_Lab> & 'C:\Program Files\Java\jdk-25.0.2\bin\j
ezwan\AppData\Roaming\Code\User\workspaceStorage\4bed134accdc63fc9436ca55b0e3d438\r
Name: Zahim Sir
Designation: Lecturer
○ PS D:\BAUST CSE\Java Programing\7th_Lab>
```

Experiment 4: Use of private access modifier

```
package Private;
public class Marks {

    private int marks;

    public void setMarks(int m) {
        marks = m;
    }
    public int getMarks() {
        return marks;
    }
}

package Private;
public class Result extends Marks {

    public void show() {
        System.out.println("Marks: " + getMarks());
    }
}

package Private;
public class Main {

    public static void main(String[] args) {
        Result r = new Result();
        r.setMarks(85);
        r.show();
    }
}
```

Experiment 4 output:

```
● PS D:\BAUST CSE\Java Programing\7th_Lab> & 'C:\Program Files\Java\jdk-25.0.2\
ezwan\AppData\Roaming\Code\User\workspaceStorage\4bed134accdc63fc9436ca55b0e3d
Marks: 85
○ PS D:\BAUST CSE\Java Programing\7th_Lab>
```

5. Discussion & Conclusion:

In this lab, we learned the concepts of private access modifier, instance variables, variable scope, and the **this** keyword. The use of private variables helped in understanding data hiding and encapsulation, while instance variables demonstrated how each object maintains its own separate data. The concept of scope clarified where variables can be accessed within a program, and the **this** keyword helped identify the current object and resolve naming conflicts between local and instance variables. Through these implementations, the understanding of object-oriented programming became stronger.

7. Reference:

1. <https://www.geeksforgeeks.org/java/encapsulation-in-java/> accessed at 5.20 PM on 7th May 2026.